

Ryan Zheng

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Mechanical Engineering

Waterloo, ON

Sep. 2021 – June 2026

EXPERIENCE

Engineering Analyst - Stamping

Sep. 2025 – Dec. 2025

Toyota Motor Manufacturing Canada

Cambridge, ON

- Eliminated a bottleneck issue with the stamping press sheet feeder with, **decreased the total line downtime by 3%**
- Led a **plant-wide steel cost reduction (SCR) project** for a steel blank that **reduces overall cost by over \$1,000,000 a year**
- **Collaborated with Toyota Motor Manufacturing Indiana to design and fabricate a hydraulic workbench** that allows parts maintenance to be done in house and **reduces maintenance time by 85%**
- Identified and organized root causes of weekly downtime by communicating with production members and reported to engineering managers

Engineering Analyst - Pilot Assembly

Jan. 2024 – Apr. 2024

Toyota Motor Manufacturing Canada

Cambridge / Woodstock, ON

- Implemented a laser locating system for the hub nut tightener that **reduces the total line downtime by 7%**
- **Designed and fabricated 8 jigs and fixtures for pre-launch vehicle assembly**, which allowed a smooth transition for updated vehicle model
- Conducted a study on the taillight temperature in cold environments to ensure the correct assembly metrics are met for the pre-launch vehicle model
- Managed large pull ahead projects such as AGV cage demolition and assembly line extension

Operation Technician

Sep. 2022 – Dec. 2022

Swap Robotics

Kitchener, ON

- Led robot assembly team, **oversaw tasks such as PCB programming, soldering, and overall robot assembly**
- **Designed and 3D Printed a custom charger solution** capable of charging 12 body cameras simultaneously while indicating charge progress
- Developed a protocol to test all robots functionalities, which sped up testing time by up to 50%

Test Lab Technician

Jan. 2022 – Apr. 2022

Multimatic Inc.

East Gwillimbury, ON

- **Tested reliability of prelaunch and in-production parts for Fords, General Motors, and Chrysler** by running tests such as ball joint/ bushing push outs, UTS tests, ball joint function fatigue tests, and torque measurements
- Led constructions of actuator setups to test reliability for control arms; conducted analysis using SOLIDWORKS to meet measurement requirements
- **Analyzed and graphed numerical data from torque measuring software** to ensure production criteria are met

PROJECTS

Precision Thermal Monitoring System | *SolidWorks, Ansys, CNC, Machine Tools*

Sep. 2025 - Apr. 2026

- Designed and deployed a weather-resistant mounting and enclosure system that allows multi-ecosystem thermal data collection for environmental monitoring
- System currently being used by Professor Roydon Fraser for grad research at the University of Waterloo Environmental Reserve

Remote Control Load Carrying Robot | *SolidWorks, Arduino, ESP32, Machine Tools*

Jan. 2025 – Apr. 2025

- Designed and built a remote control robot capable of carrying over five times its own weight

TECHNICAL SKILLS

Softwares: SolidWorks, AutoCAD, MATLAB, Ansys, CAMMaster, G-Code, Arduino, C/C++, HTML, CSS, JavaScript

Machine Tools: Comfortable and experienced with most machine tools